

MUHAMMAD JUNAID

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Faisalabad, Pakistan

PROFESSIONAL SUMMARY

Electrical Engineer with hands-on experience in end-to-end embedded firmware development, IoT system design. Proven ability to deliver production-ready solutions across ARM Cortex-A, AVR, ESP32, and Linux-based SoCs. Proficient in C/C++ and Python for real-time systems, bare metal code development, and additional skills in AI and as Service Engineer. Adept at bridging hardware and software to deliver scalable, reliable products.

PROFESSIONAL EXPERIENCE

Jr. Embedded Systems Engineer | Innovate-it, Faisalabad

Oct 2024 – Present

- Designed and developed firmware for microcontrollers from scratch, covering full product lifecycle including work flow, code flow, and device integration through to hardware assembly and testing.
- I2C, UART, and SPI, integrated diverse sensors and peripherals using communication protocols including
- Bluetooth Classic, BLE, and Wi-Fi Implemented wireless connectivity solutions including
- Built complete IoT end-to-end devices from requirements and firmware architecture through hardware bring-up, sensor integration, and final validation testing.
- Arduino AVR & ARM boards, ESP32 (dual-core Xtensa LX6), and Raspberry Pi (quad-core ARM Cortex-A72). Worked across multiple hardware platforms:
- C, C++, and Python, Developed firmware and utility scripts in
- Independently authored custom drivers and libraries when third-party solutions were unavailable, resolving complex integration, calibration, and mathematical/algorithm challenges.
- Built and maintained a personal reusable code library, significantly accelerating development cycles and enabling more time to be invested in new feature development.
- Successfully delivered 8 complete end-to-end products and contributed to 15+ projects, working largely independently with minimal supervision as a junior engineer.

Additional Roles (Supporting Capacity) | Innovate-it, Faisalabad

Oct 2024 – Present

AI Engineer (Computer Vision)

- Developed a person detection and face recognition system using YOLO for object detection as the core framework.
- Trained and deployed custom YOLO object detection models using domain-specific datasets for real-world production environments.

Service Engineer

- Provided on-site and remote support including hardware installation, device troubleshooting, maintenance, and direct client-facing customer support.

SELECTED PROJECTS

- Computer Vision People Counter:** YOLO-based real-time people counting system with custom dataset training, staff filtering logic, and edge deployment.
- Offline Crypto Hardware Wallet:** Air-gapped ESP32 wallet with cryptographic signing, secure EEPROM key storage, and QR-code transaction transfer for offline security.
- IoT Hydration Monitoring Network:** Multi-node ESP32 sensor mesh with Raspberry Pi gateway, weight sensor integration, and real-time web dashboard for analytics and alerts.
- Smart Digital Lock System:** BLE-enabled secure lock with mobile app integration, anti-tamper detection, audit logging, and scalable fleet management architecture.
- RFID Attendance Terminal:** Cloud-connected RFID system with API validation, dual BLE/Wi-Fi provisioning, and automated attendance logging with reporting dashboard.
- Smart Irrigation Controller:** Programmable embedded controller with soil moisture sensing, short-circuit protection, scheduled automation, and OLED status display.
- GSM-Integrated MCU Communication System:** Embedded system combining GSM module with MCU for remote SMS alerts, voice call triggering, and real-time field status reporting.
- Human Presence Detection via Microwave Radar:** Non-visual occupancy detection using microwave radar integrated with embedded firmware for reliable human presence sensing in low-light or obstructed environments.

EDUCATION

B.Sc. Electrical Engineering — NFC-IEFR, affiliated with UET Lahore

2020 – 2024

INTERSHIPS

Research Fellow – Cardiac ICU | IMP Scandinavian, Denmark

Dec 2024 – Jan 2025

- Conducted device testing and clinical data analysis for ICU cardiac patients to support optimization of medical monitoring equipment.

Electrical Engineering Intern | Harbin Electric Pvt Ltd, Karachi

Apr 2023 – Jun 2023

- Assisted with installation, testing, and preventive maintenance of high-voltage electrical systems under senior engineer supervision.

IoT Internship | NFC-IEFR, Faisalabad

Jul 2022 – Aug 2022

- Built embedded IoT prototypes using Arduino Uno, integrating Wi-Fi, Bluetooth, and sensor modules for hands-on system design training.

TECHNICAL SKILLS

Languages: C, C++, Python **Hardware** Arduino (AVR & ARM), ESP32 (Xtensa LX6), Raspberry Pi (ARM Cortex-A72) **Protocols:** UART, SPI, I2C, BLE, Wi-Fi, API Integration **Domains:** IoT Device Development, Embedded Firmware, Sensor Integration, Hardware Bring-up & Testing **OS/Tools:** Ubuntu/Linux, Windows, VS Code, PyCharm, Arduino IDE **Languages:** English (Intermediate), Urdu (Native)